

Lecture 6

Extended Entity Relationship (E-R) Modeling

Crow's Foot Model vs. Chen Model in Handling the Cardinality Constraints

Tape Rental Conceptual Model

Crow's Foot Model



Customer must rent one or more tapes.
Tapes may be rented by zero or more customers.

Chen Model



Decomposition of the M:N Relationship

Crow's Foot Model

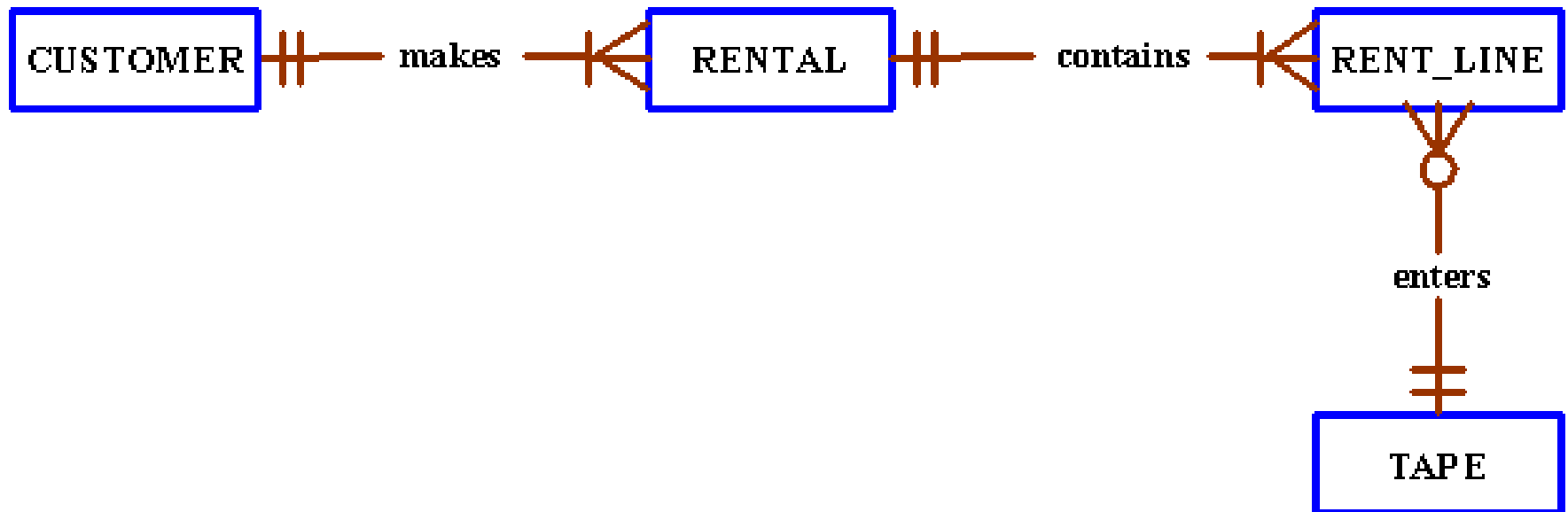


Chen Model



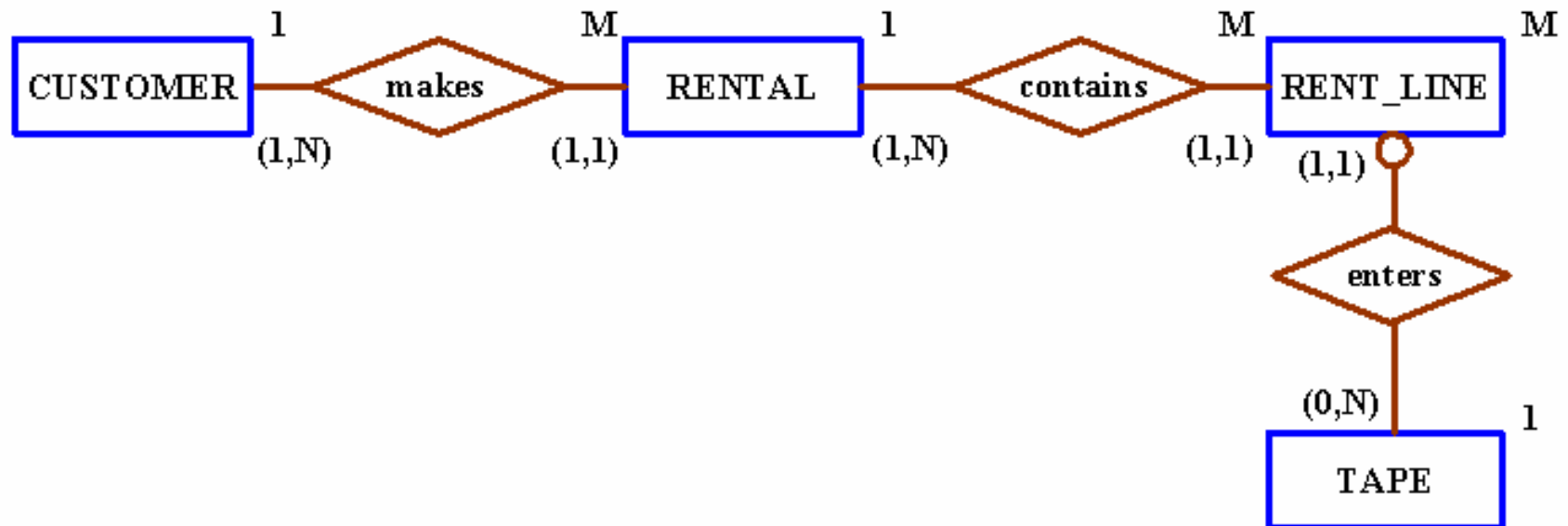
The Updated Tape Rental Database Design

Crow's Foot Model



The Updated Tape Rental Database Design (cont.)

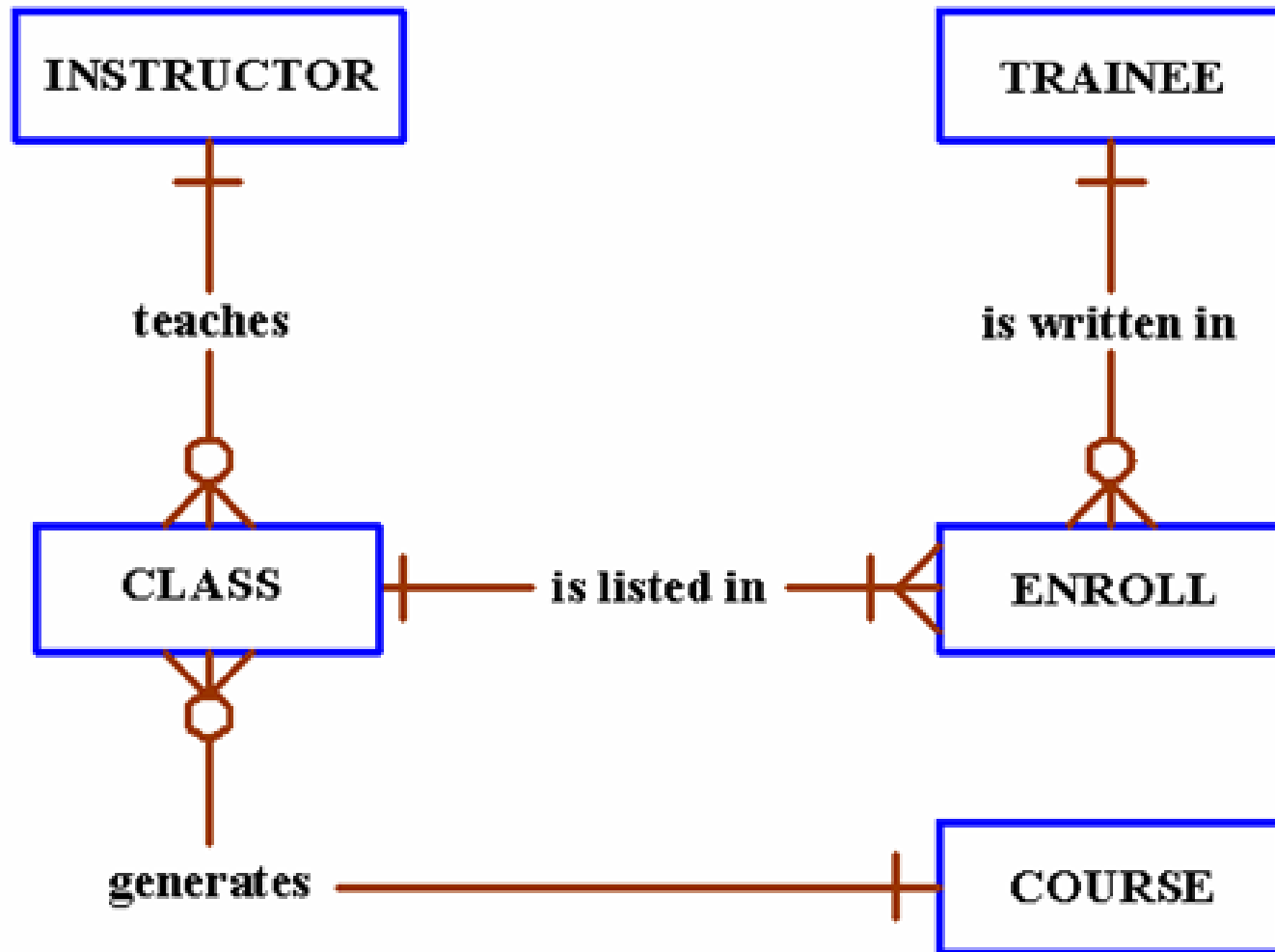
Chen Model



The Chen and Crow's Foot E-R Diagrams for HEG

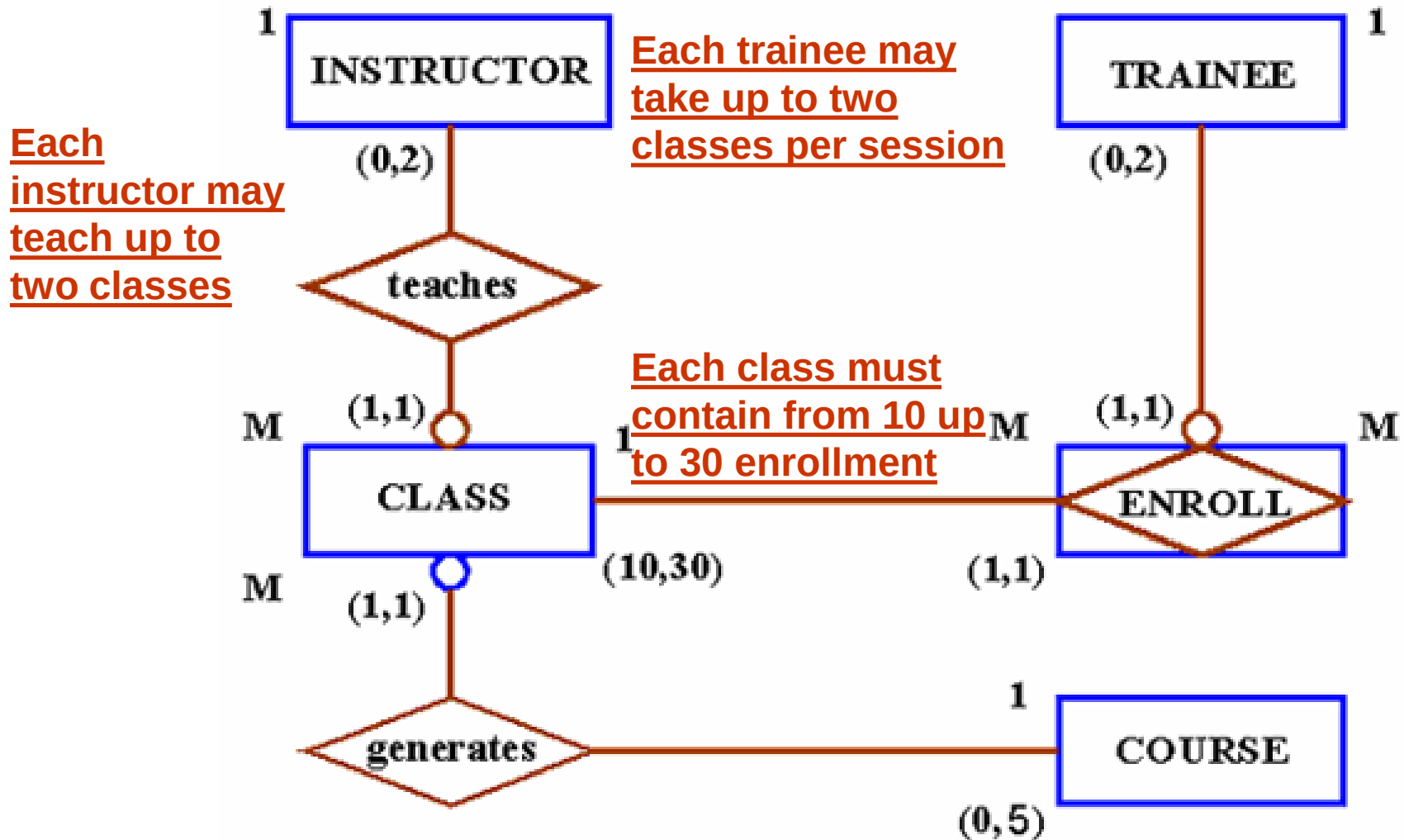
“The HEG has 12 instructors and can handle up to 30 trainees per class. HEG offers five "advanced technology" courses, each of which may generate several classes. If a class has fewer than 10 trainees in it, it will be canceled. It is, therefore, possible for a course not to generate any classes during a session. Each class is taught by one instructor. Each instructor may teach up to two classes or may be assigned to do research. Each trainee may take up to two classes per session.”

HEG: Crow's Foot Model



HEG: Chen Model

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There are five "advanced technology" courses and it is possible for a course not to generate any classes

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Extended Relational Analysis

Subsets of Entities
Dependents of Entities
Recursive Relations
Complex Relations
Time Relations

Subsets of Entities

A subset of an entity is a collection of occurrences of an entity type that participate in relations or possess attributes that are not common to the entity type as a whole. The entity as a whole is called a superset in this context.

- Subsets are nouns.
- Subsets can participate in relations.
- Subsets can have attributes.

For example, when you ask: "Is this a relation between Persons and Department?", the user will reply: "Yes, but only the employees"

EMPLOYEE

EMPLOYEE NUMBER	DEPT CODE	EMPLOYEE NAME	SALARY AMOUNT	HOURLY AMOUNT
PK, SA	FK	NN		
1	SA	JONES	15,000	
2	PA	SMITH	10,000	
3		ADAMS		12.50
4		WHITE		12.75
5	PA	BROWN	12,000	

All employees have names, but only salaried employees have departments and salaries. The hourly amount applies only to non-salaried personnel. This results in a significant number of nulls, and provides no clear distinction between employee types.

Classifying Salaried Employees and Hourly Employees as subsets of Employee, however, results in a somewhat different model:

EMPLOYEE

EMPLOYEE NUMBER	EMPLOYEE NAME
PK, SA	NN
1	JONES
2	SMITH
3	ADAMS
4	WHITE
5	BROWN

The PK of Subset table are the same as PK of Superset table, it is also FK in the Subset table.

SALARIED EMPLOYEE

EMPLOYEE NUMBER	DEPT CODE	SALARY AMOUNT
PK, FK	FK, NN	NN
1	SA	15,000
2	PA	10,000
5	PA	12,000

HOURLY EMPLOYEE

EMPLOYEE NUMBER	HOURLY AMOUNT
PK, FK	NN
3	12.50
4	12.75

Example

Here we have two separate entities, Customer and Employee. Note that there is a person named Jones in each table, with two different primary key values.

If this Jones is the same person in each table, we may want to consider Customers and Employees subsets of a larger entity, say, Persons, and model the situation

CUSTOMER

CUSTOMER NUMBER	CUSTOMER NAME	CREDIT AMOUNT	DUNS NUMBER
PK, UA	NN	NN	ND
123	BOYLE	10,000	98765432
234	GRANT	50,000	
345	JONES	15,000	76543210

EMPLOYEE

SOC SEC NUMBER	EMPLOYEE NAME
PK, UA	NN
123456789	JONES
234567890	SMITH
345678901	ADAMS

PERSON

PERSON NUMBER	PERSON NAME
PK, SA	NN
1	BOYLE
2	GRANT
3	SMITH
4	ADAMS
5	JONES

CUSTOMER PERSON

PERSON NUMBER	CREDIT AMOUNT	DUNS NUMBER
PK, FK	NN	ND
1	10,000	98765432
2	50,000	
5	15,000	76543210

EMPLOYEE PERSON

PERSON NUMBER	SOC SEC NUMBER
PK, FK	ND
3	123456789
4	234567890
5	345678901

Dependent of Entities

A dependent of an entity is a noun that may have **attributes** of its own but that **exists only as part of**, or **as subordinate to**, some other entity. The entity in this context is called the **parent** of the dependent.

- Dependents are **nouns**.
- Dependents can **participate in relations**.
- Dependents can **have attributes**.

Example

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BUILDING

BUILDING CODE	BUILDING NAME
PK, UA	NN
A	MAIN
B	BRANCH
C	ANNEX

BUILDING, FLOOR

BUILDING CODE	FLOOR NUMBER	SQUARE FOOTAGE
PK, FK	PK, UA	NN
A	1	50,000
A	2	10,000
B	1	12,500

BUILDING, FLOOR, ROOM

BUILDING CODE	FLOOR NUMBER	ROOM NUMBER	RENTAL AMOUNT
PK, FK, FK1	PK, FK1	PK, UA	NN
A	1	01	1,500
A	2	01	1,000
B	1	02	1,250

Dependent of Entities (cont.)

- Dependent **must have** parent.
- Dependent **must have only** one parent.
- Dependent **must always have** the same parent.
- The PK of a dependent is always the PK of its parent(s) **plus** one additional column.

Recursive Relations

- A recursive relation is a relation between occurrences of a particular entity type and other occurrences of **the same entity type**. The entity could be simple, subset or dependent entity.
- Recursive relations can be of the three basic types: 1-1, 1-M, M-N.
- Also, called **Unary Relations**.

Example

An imaginary company employs five individuals: Baker, Hill, Edison, Carnegie and Nixon. Nixon is the President and runs the show. Carnegie and Edison are Vice Presidents of Sales and Engineering, respectively. Hill and Baker are professional Salespersons, both reporting to Carnegie.

Translating to tables, we have:

EMPLOYEE		POSITION			
EMPLOYEE NUMBER	EMPLOYEE NAME	POSITION NUMBER	EMPLOYEE NUMBER	RPTS TO POSITION NUMBER	POSITION NAME
PK, SA	NN	PK, SA	FK	FK	NN
1	BAKER	1	5		PRESIDENT
2	HILL	2	4	1	UP SALES
3	EDISON	3	3	1	UP ENGR
4	CARNAGIE	4	2	2	SALESPERS
5	NIXON	5	1	2	SALESPERS

Example

A certain city has within it a number of relatively straight, uniquely named streets running in various directions. The user desires to model each of these streets, and their intersections (the points where any two streets cross).

A portion of the city map appears below, together with a data model of this situation.

	MAPLE	
FIRST	OAK	SECOND

STREET

STREET NAME
PK, UN
FIRST
SECOND
MAPLE
OAK

INTERSECTION

STREET NAME	CROSSES STREET NAME
PK, FK	PK, FK
FIRST	MAPLE
FIRST	OAK
SECOND	MAPLE
SECOND	OAK

Complex Relations

- A complex relation is a relation between **more than two entities**.

Exercise 14, page 47

The user says...

"Almost 200 employees in the north region, remember? And nearly 100 in the south. That's right, each employee works in only one region..."

"Now about the customers: over fourteen new ones in the western region alone... What? No, if they have branches in more than one region we treat 'em like separate customers - different numbers and everything..."

"The boss seems to like his new class codes. He's just about done classifying the customers (one class per customer), and next week he says he'll start on the employees. I'm qualified for most everything - banks, hospitals, the works - so I don't have to worry..."

The entities are...

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CLASS

CLASS CODE
PK, UA
H
B
G

CUSTOMER

CUSTOMER NUMBER
PK, SA
1
2
3

EMPLOYEE

EMPLOYEE NUMBER
PK, SA
1
2
3

REGION

REGION CODE
PK, UA
N
S
E

Region(1):Customer(M)+Region/Customer(M):Class(1)
=
1:M:1

CLASS

CLASS CODE
PK, UA
H
B
G

CUSTOMER

CUSTOMER NUMBER	CLASS CODE	REGION CODE
PK, SA	FK, UA	FK, UA
1	H	N
2	H	S
3	B	S

REGION

REGION CODE
PK, UA
N
S
E

Example

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COURSE	STUDENT	GRADE
COURSE NUMBER	STUDENT NUMBER	GRADE CODE
PK, SA	PK, SA	PK, UA
1	1	4
2	2	3
3	3	2

Course(N):Student(M)

+

Course/Student(M):Grade(1)

=

N:M:1

COURSE/STUDENT		
COURSE NUMBER	STUDENT NUMBER	GRADE CODE
PK, FK	PK, FK	FK
1	1	4
1	2	3
2	1	4

Example

A company processes orders for various kinds of parts which are purchased from outside sources called vendors. Orders, parts and vendors are all identified with system-assigned keys. Each kind of part can be supplied by more than one vendor; the part number, however, remains constant. Each vendor can also supply more than one kind of part. Orders specify not only the required kinds of parts and the quantity, but the desired source - that is, the vendor who is to supply those parts on that order.

ORDER

ORDER NUMBER
PK, SA
1
2
3

PART

PART NUMBER	PART NAME
PK, SA	NN
1	SPARK PLUG 1A
2	INFANT DIAPERS
3	10W40 OIL

VENDOR

VENDOR NUMBER	VENDOR NAME
PK, SA	NN
1	CHAMPION
2	J & J
3	ACME ENG

PART/VENDOR

PART NUMBER	VENDOR NUMBER
PK, FK	PK, FK
1	1
1	3
2	2
3	1

ORDER/PART/VENDOR

ORDER NUMBER	PART NUMBER	VENDOR NUMBER	QUANTITY
PK, FK	PK, FK, FK	PK, FK, FK	NN
1	1	1	100
1	1	3	100
2	3	1	200
2	2	2	100

Part(M):Vendor(N)+**Part/Vendor(M):order(N)**

=

M:N:M

Time Relations

A Time Relation is a relation between an entity, a subset, a dependent, or another relation and time.

- Time Relations are always many-to-many.
- Imagine a relation between Years, Months, Days, Hours and Seconds.
- We only consider a virtual entity called **TIME**.

TIME	
POINT IN TIME	
YYYY/MM/DD	HH:MM:SS
PK, SA	
...	
1953/07/15	23:20:13
...	

Example

The user is manager of a high school football team and wishes to track player weights over time. Players have non-unique but mandatory names and weigh in daily.

Translating to tables, we have:

PLAYER

PLAYER NUMBER	PLAYER NAME
PK, SA	NN
1	BRUISER
2	ANIMAL
3	MAD DOG
4	BULL
5	BAMBI

PLAYER HISTORY

PLAYER NUMBER	YY/MM/DD	WEIGHT
PK, FK	PK, FK	NN
1	87/11/22	265
1	87/11/23	265
1	87/11/24	310
2	87/11/22	215
2	87/11/23	215

Example

A particular organization is divided into departmental units and its employees are assigned to only one department at a time. The user, however, wishes to keep a record of previous departmental assignments, and also desires to insert future transfers into the model that will automatically take effect on the specified day.

Translating to tables, we arrive at the model below.

DEPARTMENT	EMPLOYEE	EMPLOYEE/DEPARTMENT/HISTORY		
DEPT CODE	EMPLOYEE NUMBER	EMPLOYEE NUMBER	EFFECTIVE YY/MM/DD	DEPT CODE
PK, UA	PK, SA	PK, FK	PK, FK	FK
SA	1	1	87/01/01	SA
PA	2	1	87/07/15	PA
RE	3	1	99/01/01	RE
MK	4	2	87/01/01	PA
GR	5	2	87/01/15	